

German University in Cairo

Mechatronics Lab(MCTR704)

Colour Sorting Mechanism

Project No. [26]

Group: 31

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Project Description

• Project Title: Color Sorting

Objective:

The main objective of this pneumatic mechanical system is to sort and categorize two different items based on their color.

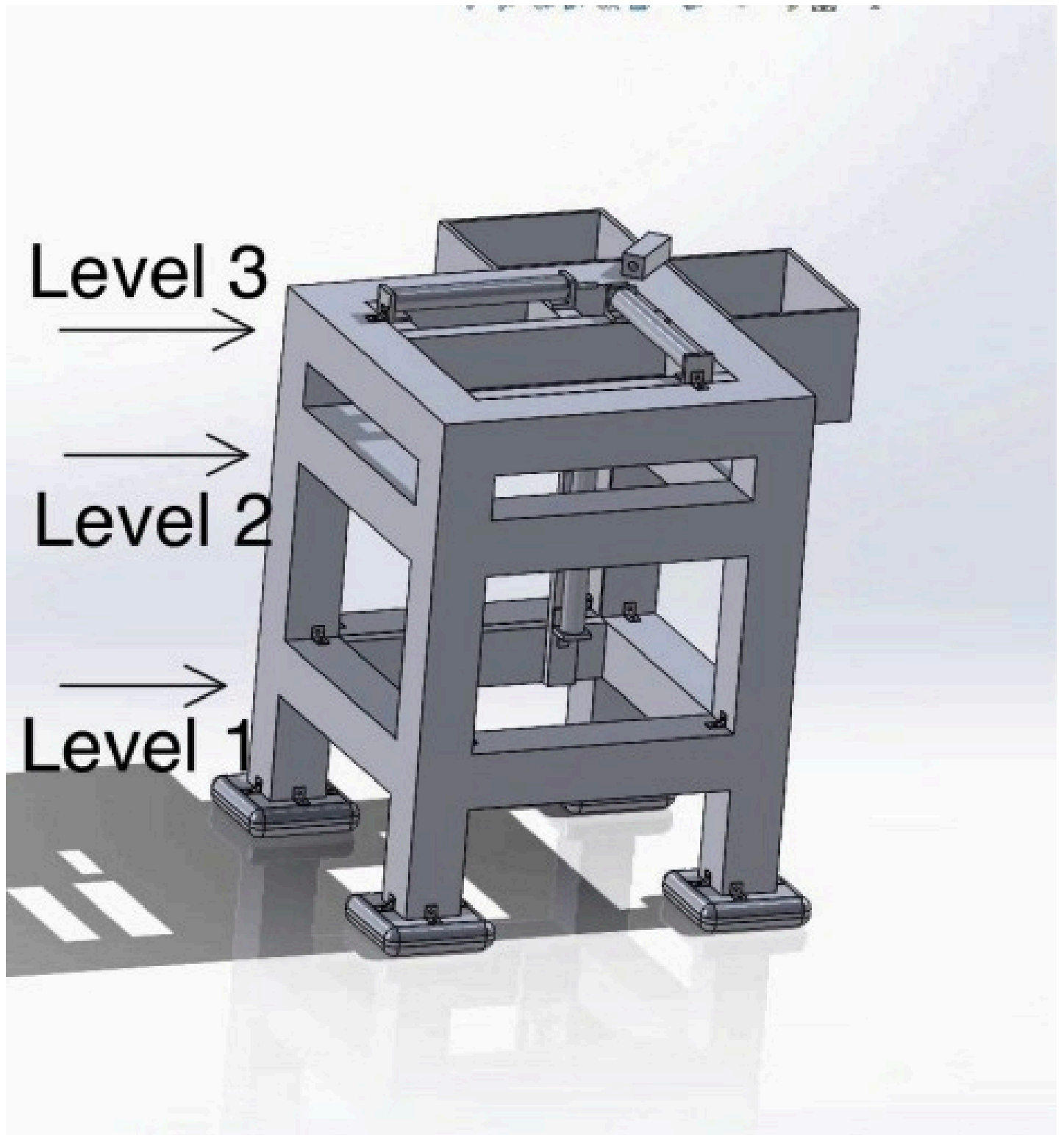
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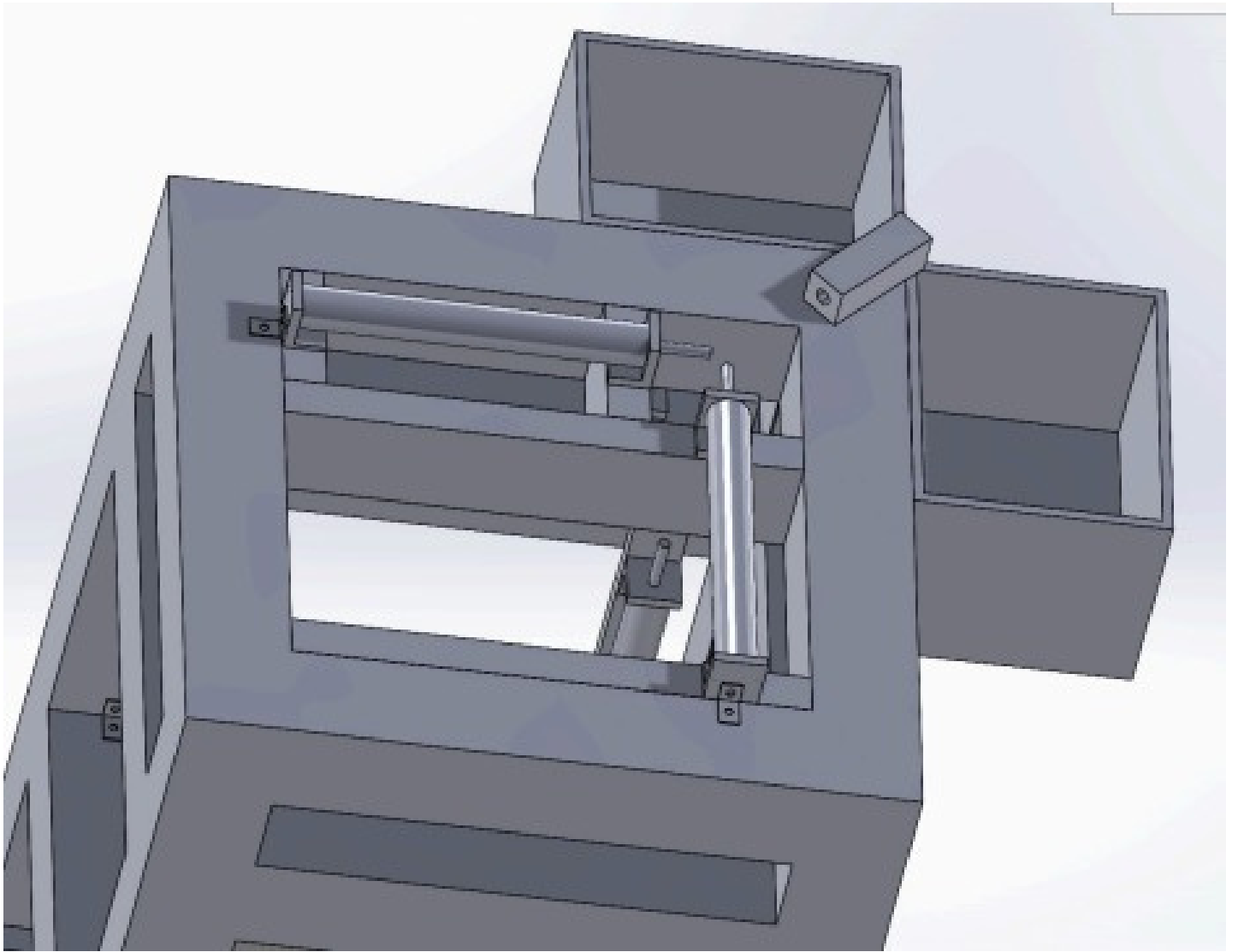
- The sorting system consists of a ground-mounted frame structure.
- The frame supports the mechanical components of the sorting system.
- The system includes three pneumatic cylinders.
- The first cylinder, referred to as the "main cylinder," receives the item to be sorted and lifts it to the sensor-based decision platform.
- Based on the sensor detection, one of the remaining two cylinders extends to push the work piece to its designated location.
- For light-colored items, Cylinder 2 extends to guide the work piece to its assigned position.
- For dark-colored items, Cylinder 3 extends to direct the work piece to the alternate location.

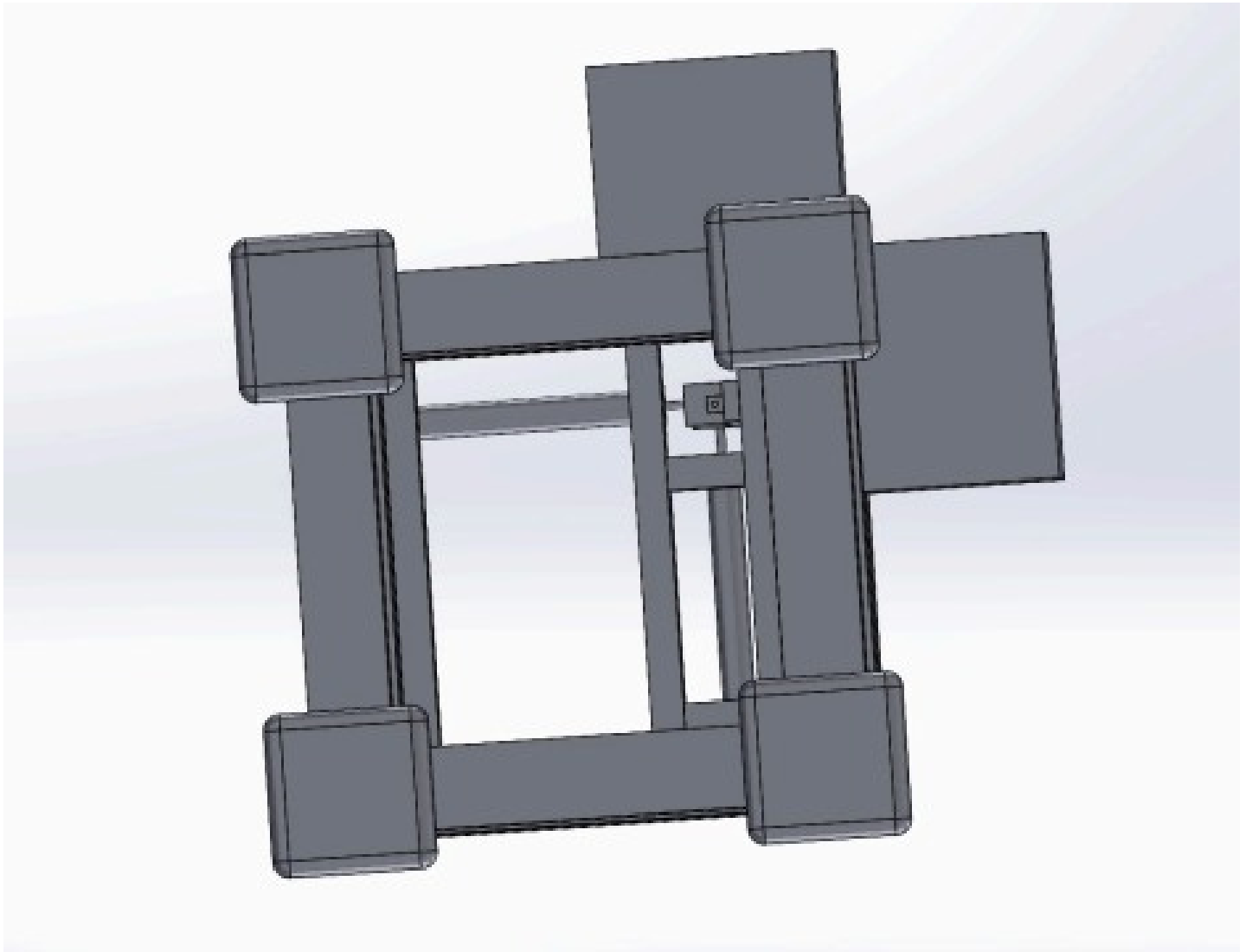
The description above outlines the core operation of the system. Complete your design to ensure the process operates continuously through multiple cycles. Identify and include all necessary components required for proper system operation, such as actuators, sensors, and control elements.

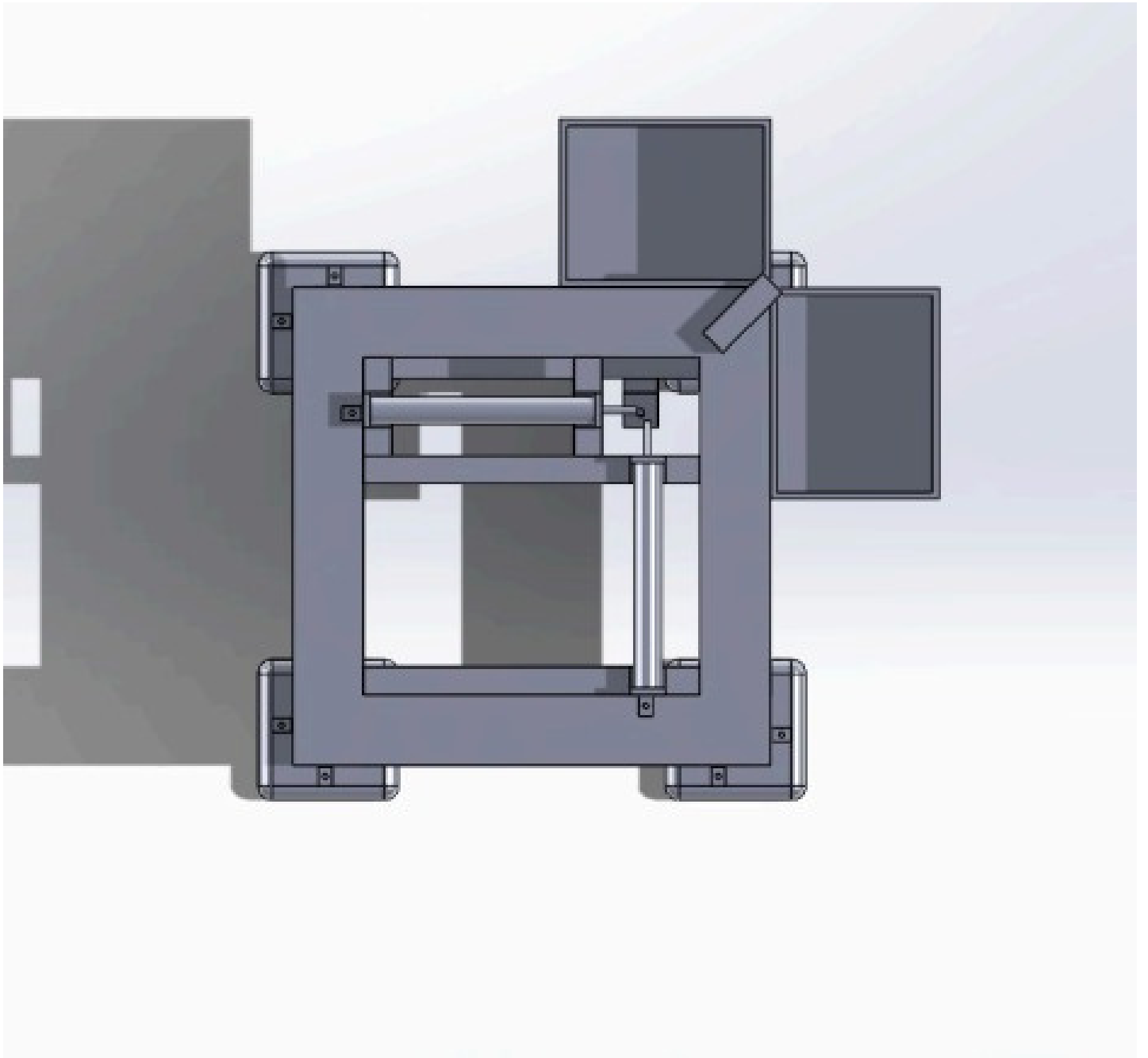


Solid works Design: 3D Schematic Diagram

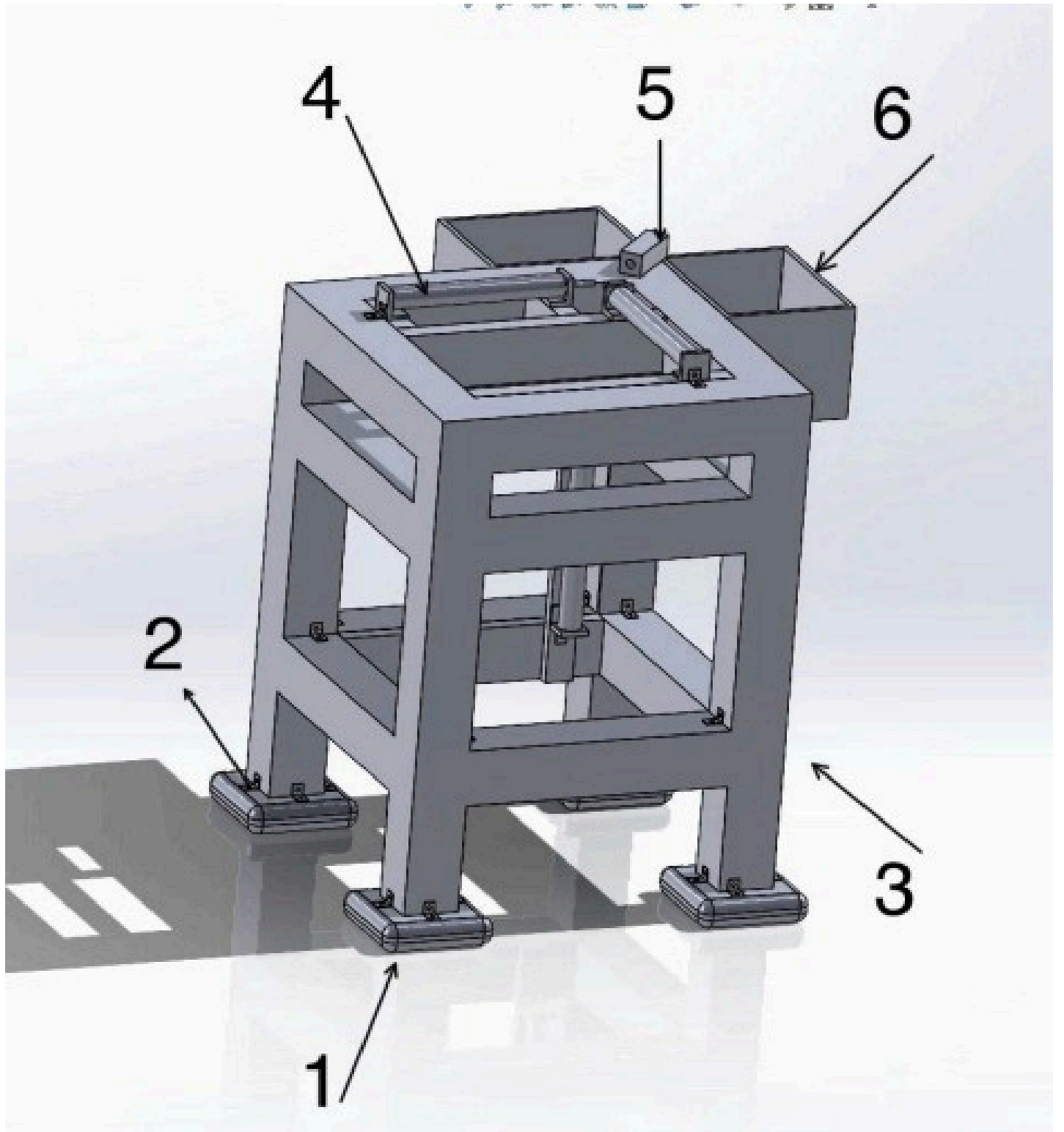






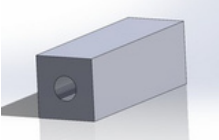
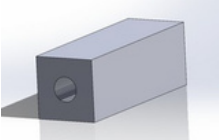


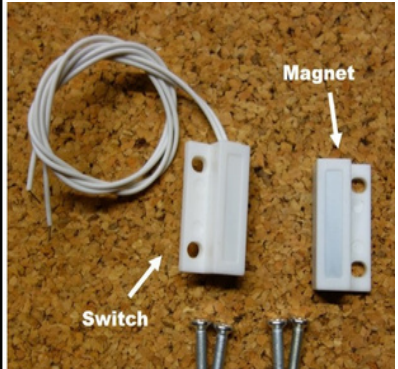




Part Number	Name
1	Base Plate
2	Mounter (L shaped)
3	Main Table Frame (composed of beams/rods)
4	Pneumatic Cylinder + Piston + End Effector
5	Colour Sensor + PhotoElectric Presence sensor
6	Sorting Boxes (2)

Project Components List and Data Sheets Description:

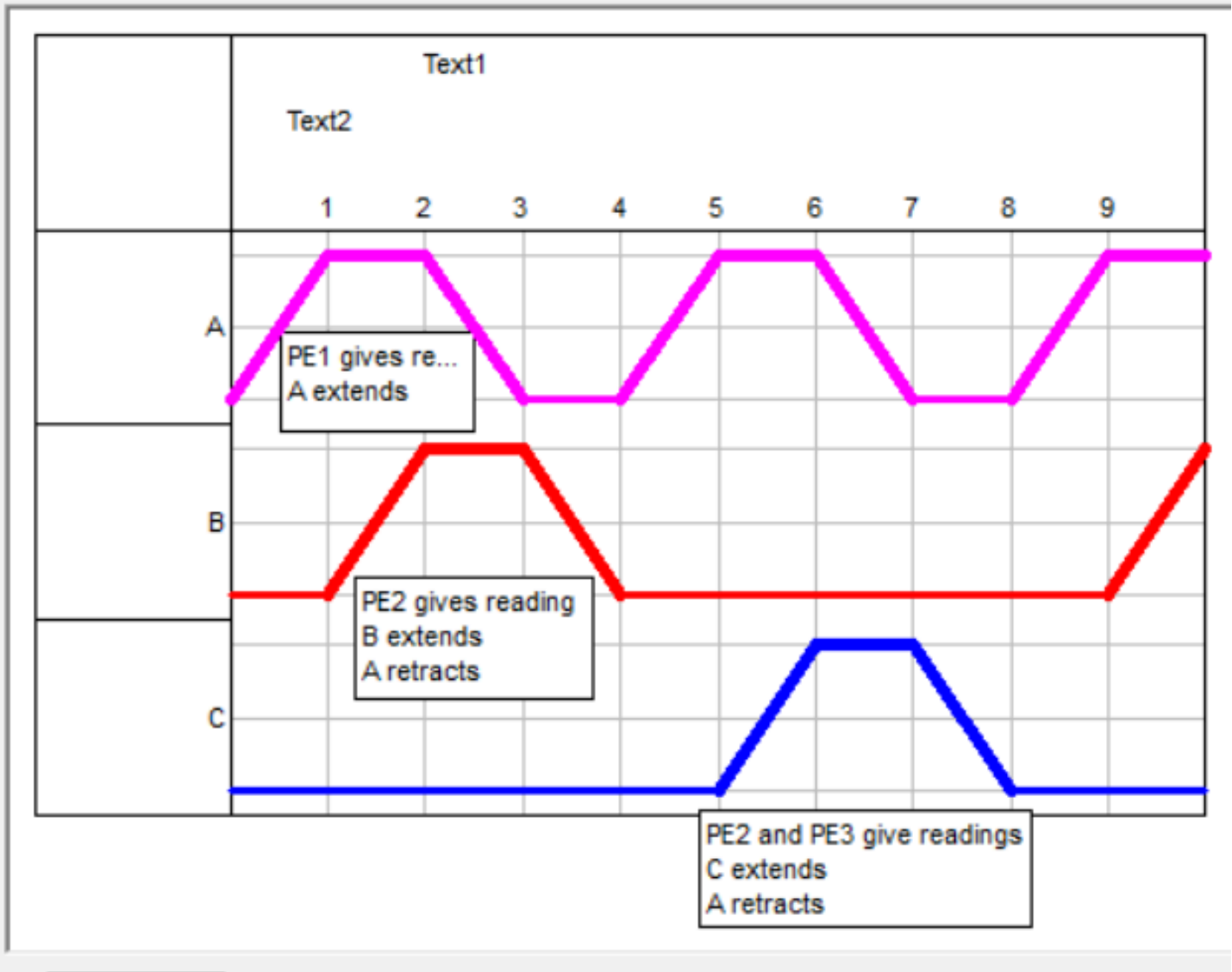
	Name	Description	Quantity	Photo
1	Pneumatic Cylinder	Cylinders that displace the object to the desired position in the mechanism	3	
2	Photoelectric sensor	Detects presence of	1	
3	Colour sensor	Outputs data that represents the colour of the object	1	
4	Relay with base 24VDC 4 Switching Groups	A relay is an electrical switch controlled by a small signal. It lets a low-power controller safely turn high-power devices (like solenoid valves) ON and OFF.	6	

5	Reed Switch	It detects whether a pneumatic cylinder is fully extended or fully retracted by sensing the magnet inside the piston.	6	
6	5/2 Solenoid Valve	It is used to control the extension and retraction of a double-acting pneumatic cylinder.	3	
7	Photo Electric sensor	It is used to detect presence, position, materials such as plastic, glass, liquids, powders, or metal.	1	

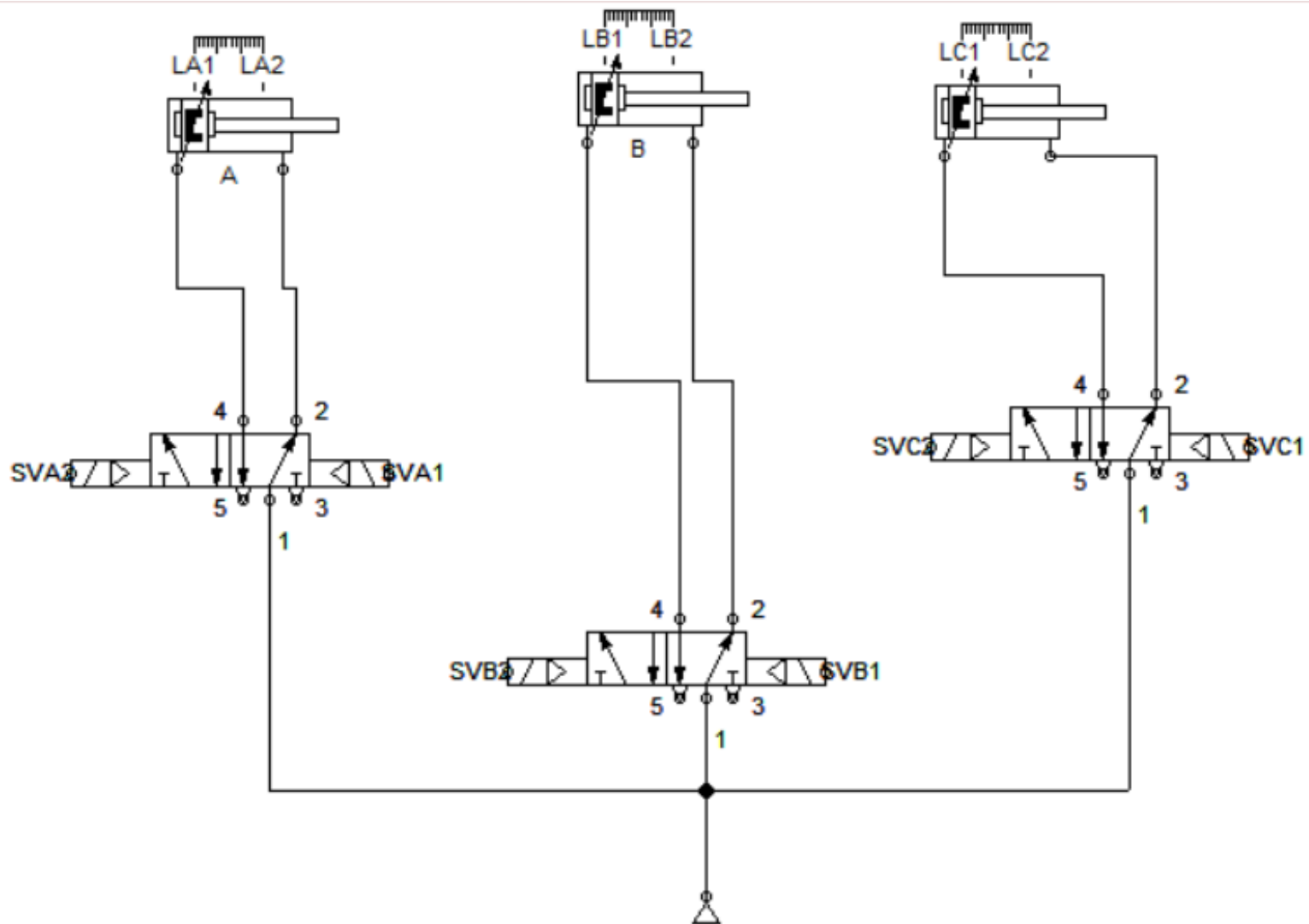
8	Copper Cables	Deliver electrical power or signals to control the pneumatic system.		
9	Start / Stop PushButtons	Start Button: Initiates the automated can crushing cycle. When pressed, the system begins operating the cylinders in sequence. Stop Button: Immediately halts the system for safety or emergency, stopping all cylinder movement and solenoid activation.	2	

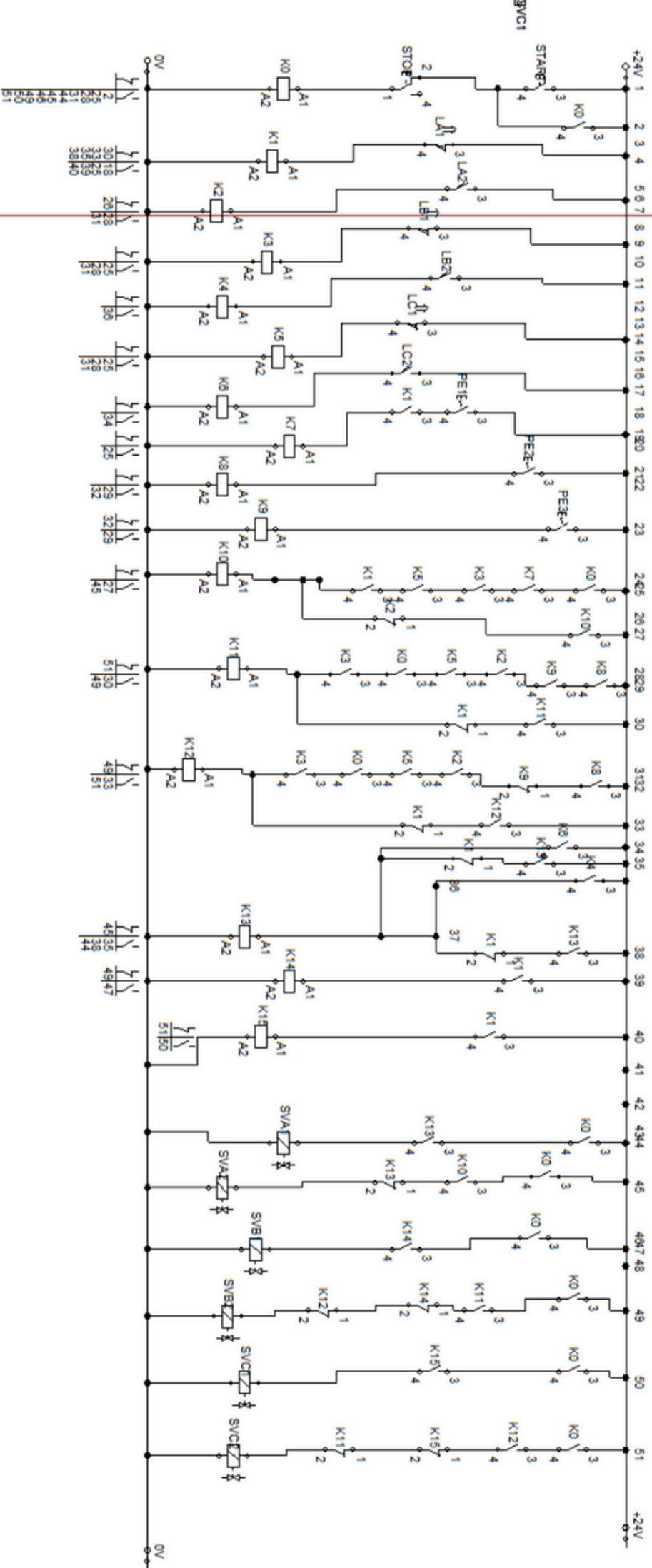
10	Power Supply	24V DC Power Supply Unit (PSU) that takes the local AC mains voltage (e.g., 120V AC or 230V AC) and converts it to 24V DC with enough current capacity to run the PLC, all solenoid valve coils, and all sensors simultaneously.	1	
11	Pneumatic cord	Delivers compressed air from control valves to cylinders, enabling smooth operation of tasks like pushing, crushing, and ejecting in the can crusher		

Pneumatic Step Diagram:

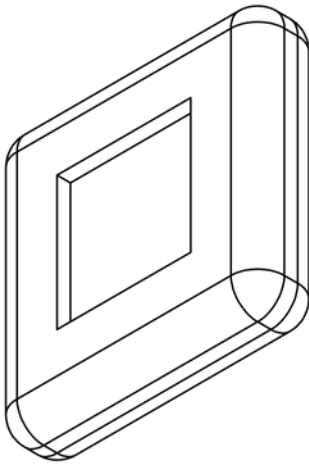
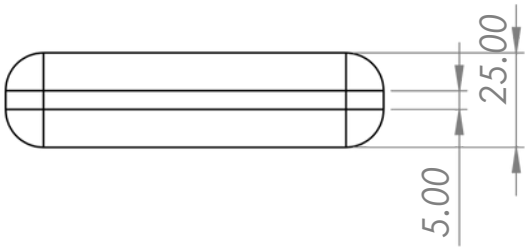
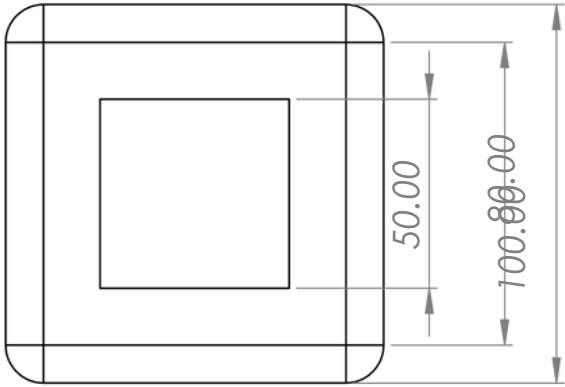


Pneumatic circuit:





Mechanical Components 2D Projections with Dimensions:



UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN MILLIMETERS
SURFACE FINISH:
TOLERANCES:
LINEAR:
ANGULAR:

FINISH:

DEBURR AND
BREAK SHARP
EDGES

DO NOT SCALE DRAWING

REVISION

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APPV'D					
MFG					
Q.A					

MATERIAL:

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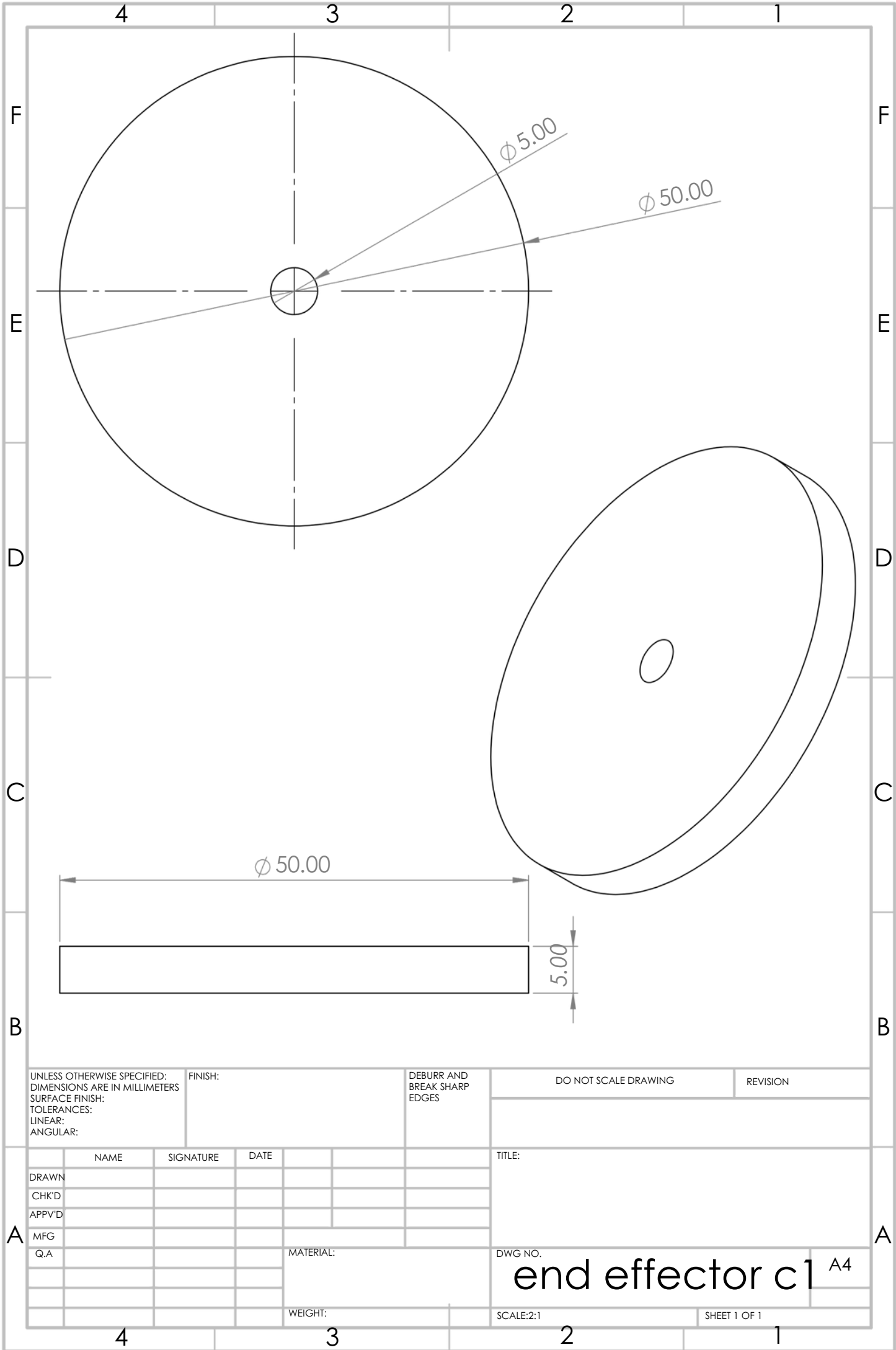
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Basefinal 1

SHEET 1 OF 1

A4



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BREAK SHARP
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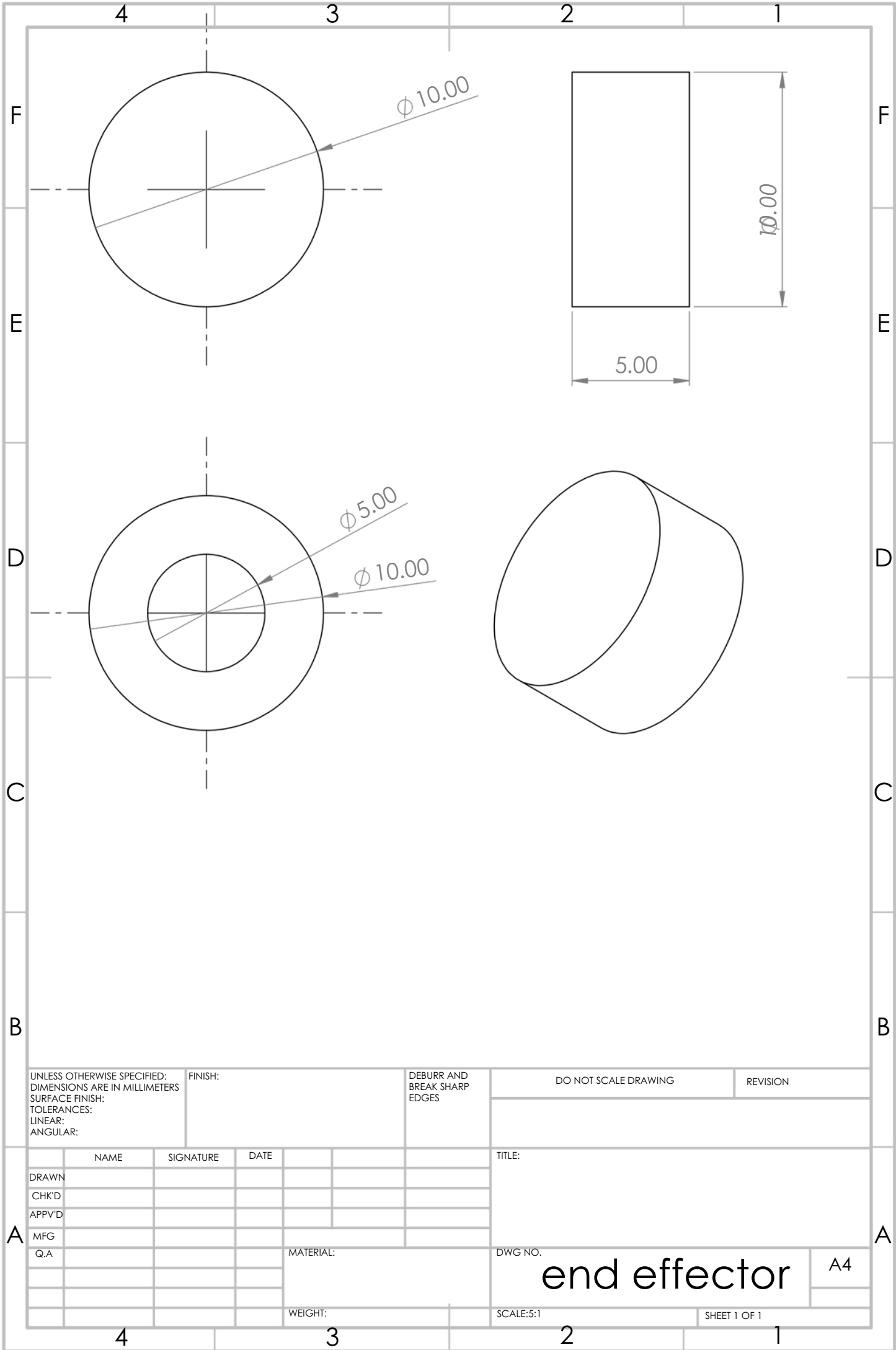
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SCALE:2:1

SHEET 1 OF 1

end effector c1 A4



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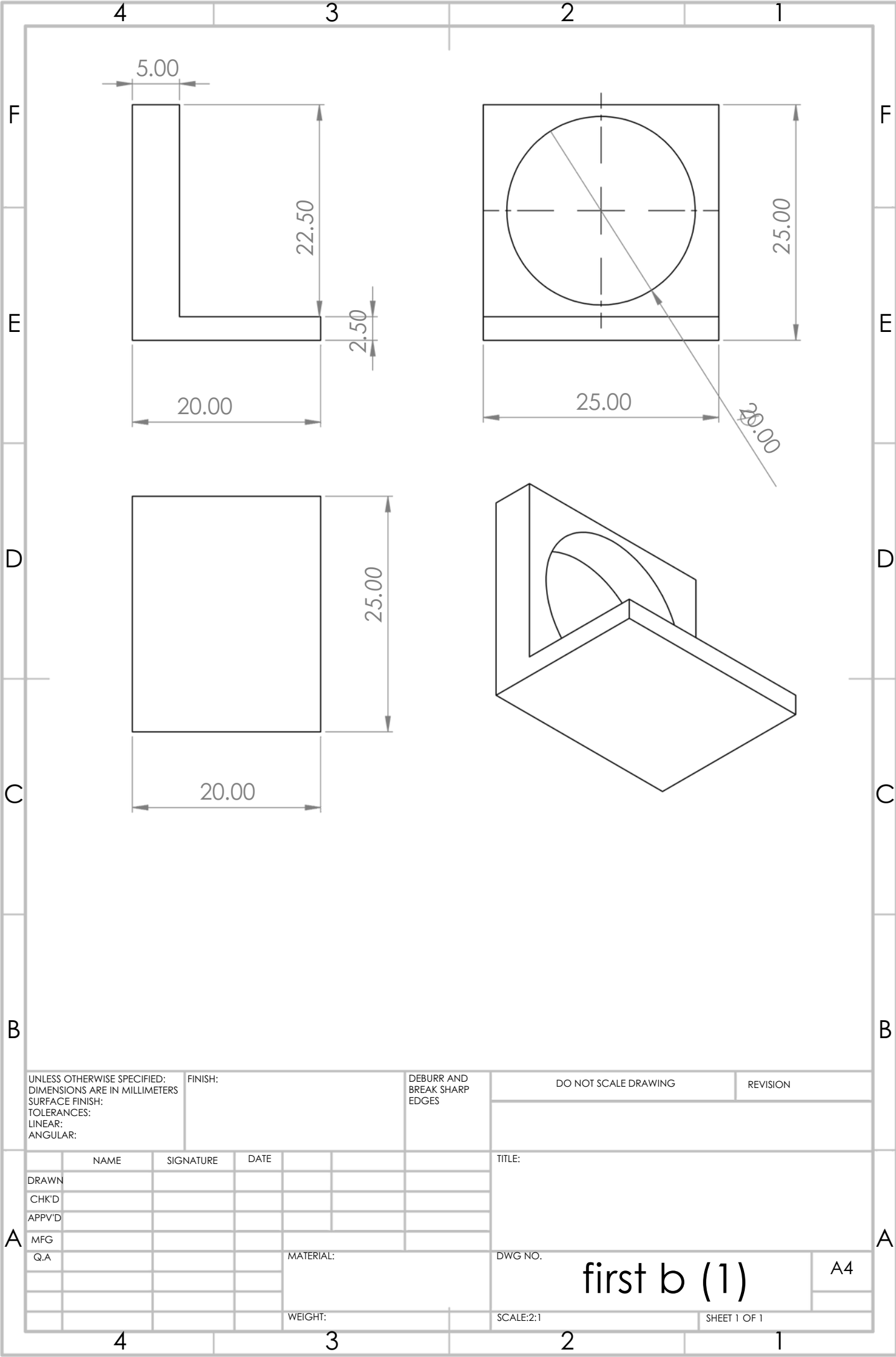
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end effector

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SHEET 1 OF 1



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BREAK SHARP
EDGES

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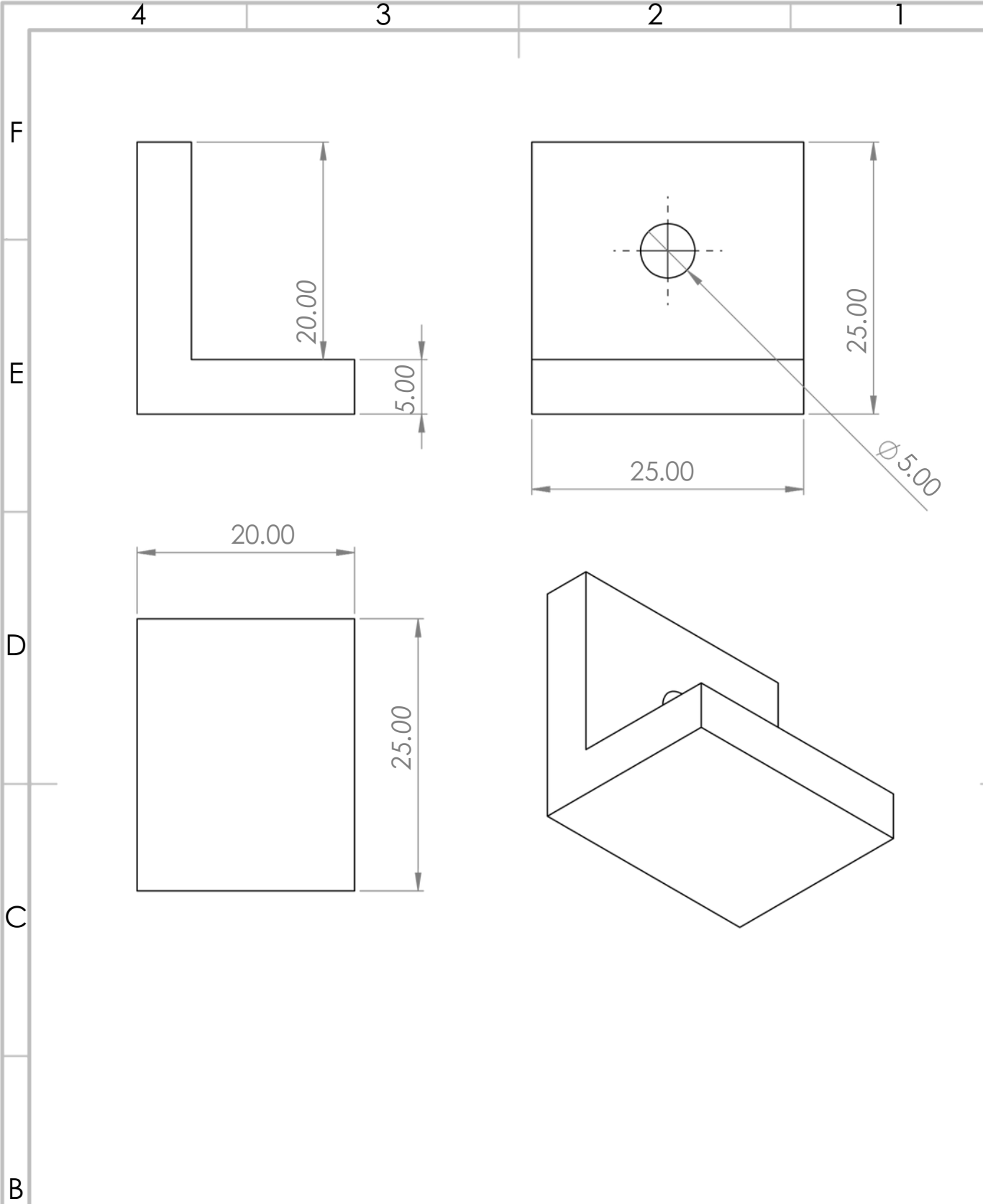
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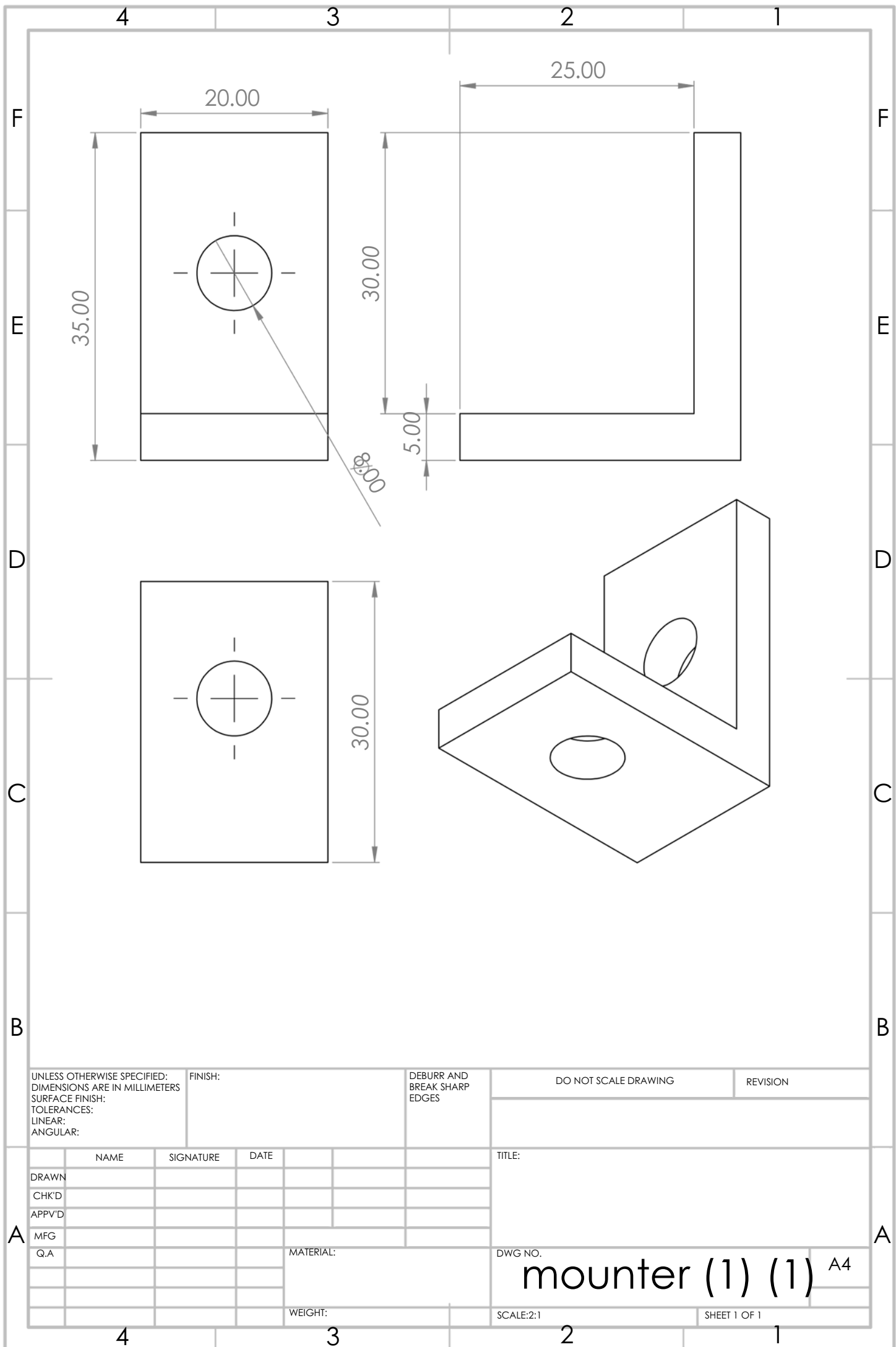
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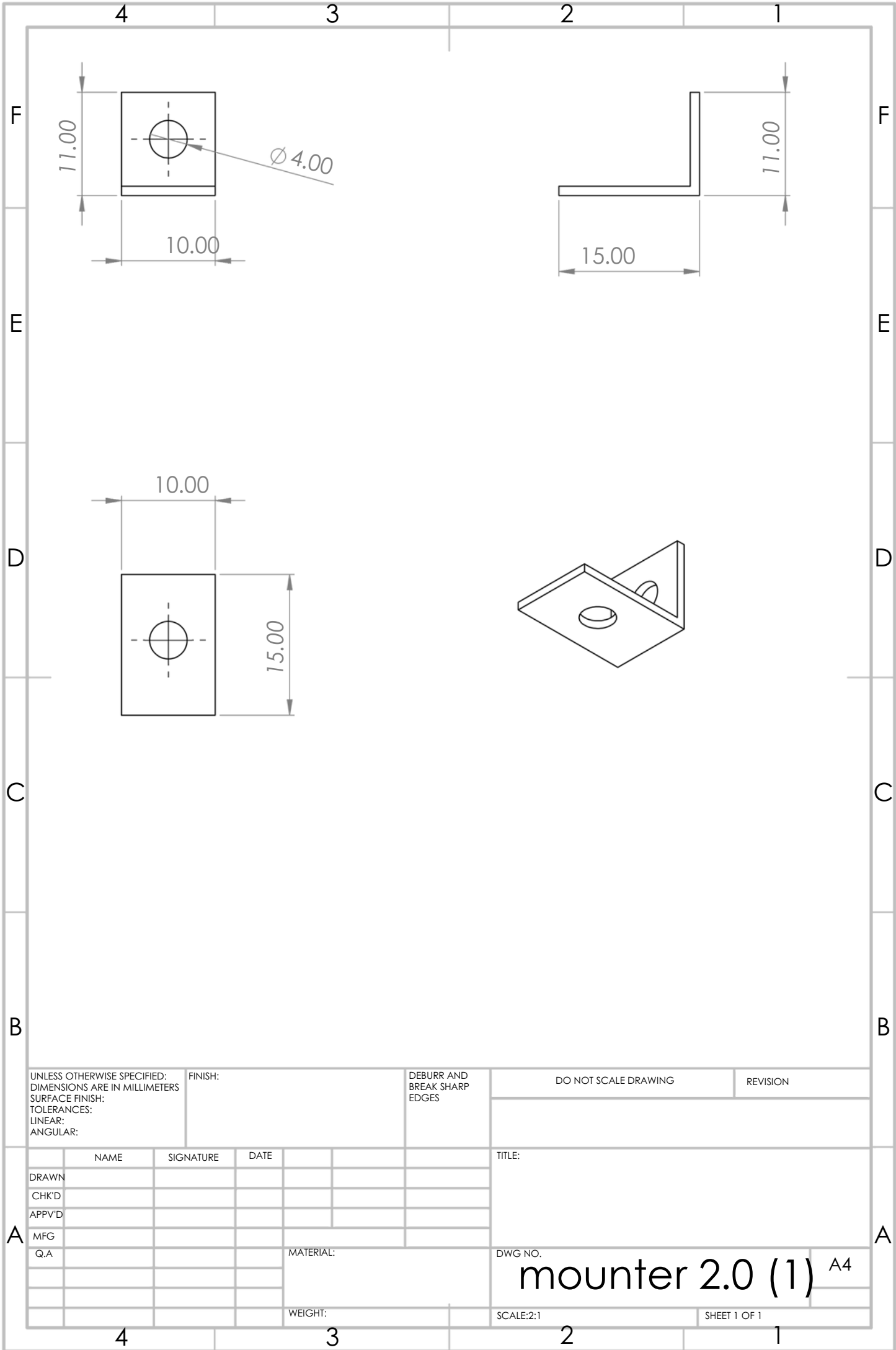
first b (1)

A4



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CHK'D											
APPV'D											
MFG											
Q.A						MATERIAL:		DWG NO.		front b' (1)	
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						WEIGHT:				SHEET 1 OF 1	





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BREAK SHARP
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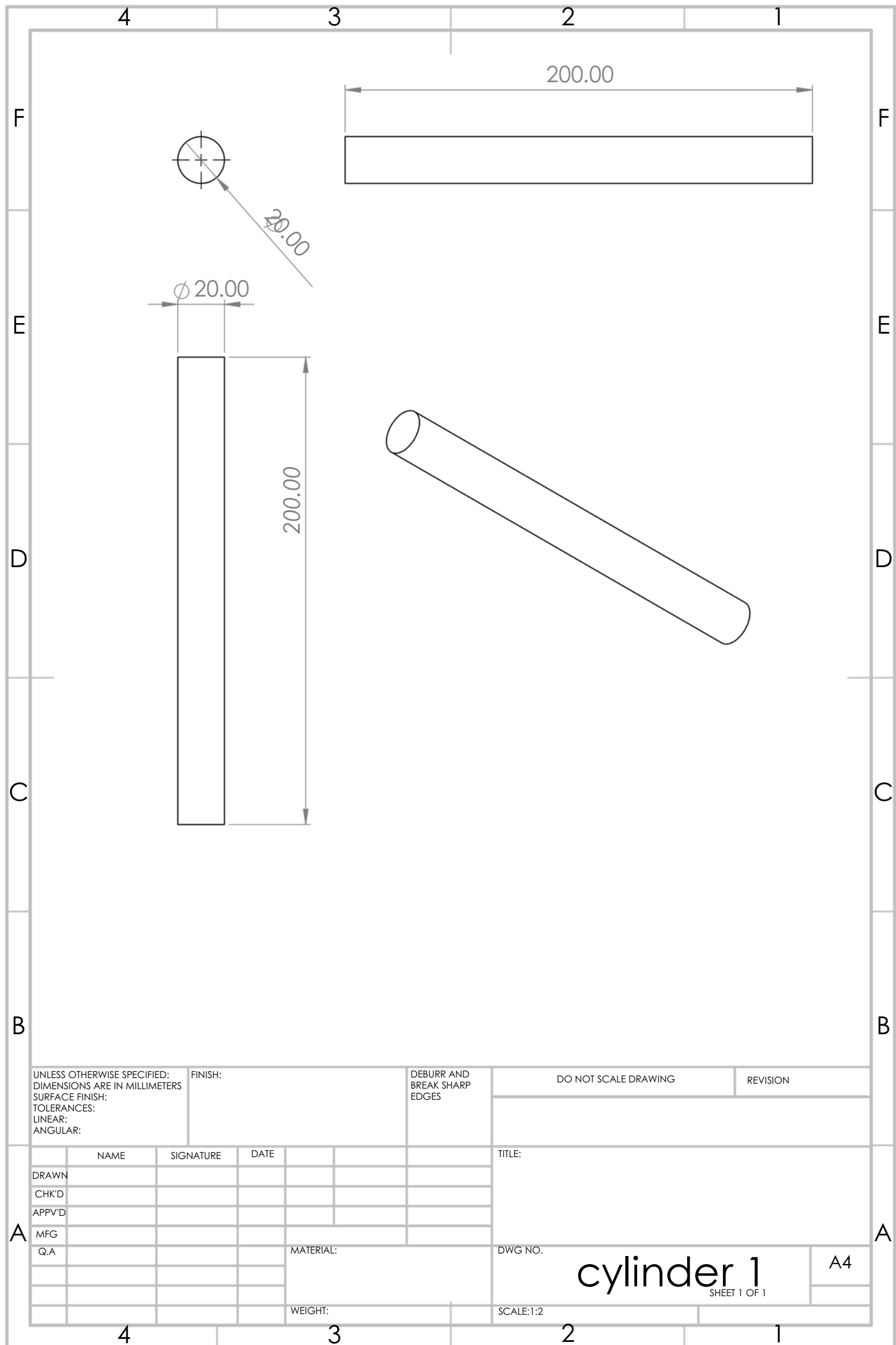
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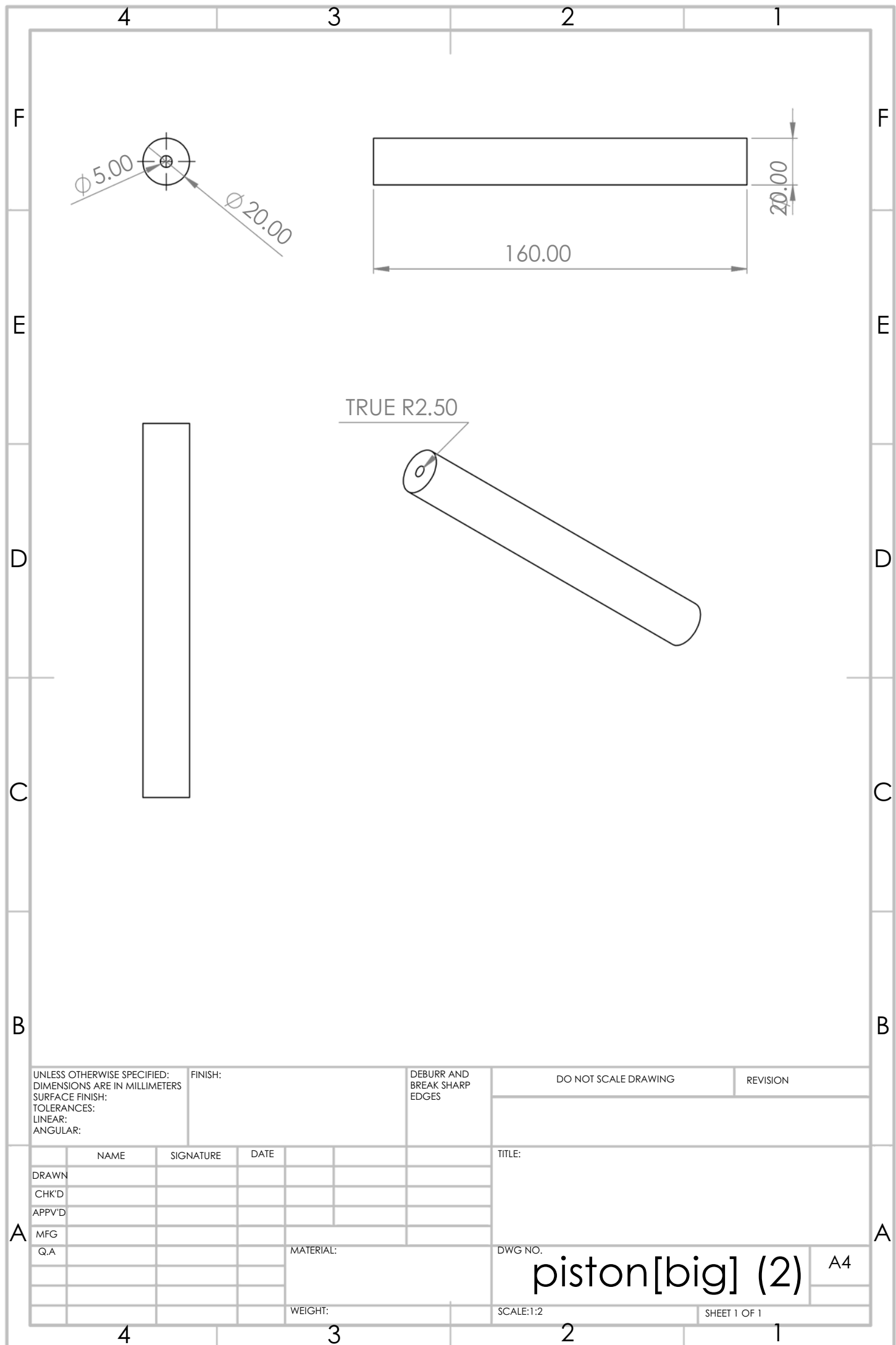
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SCALE:2:1

SHEET 1 OF 1

mounter 2.0 (1) A4





4

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TOLERANCES:
LINEAR:
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DEBURR AND
BREAK SHARP
EDGES

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SIGNATURE

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MATERIAL:

DWG NO.

rod cyl1 (1)

A4

WEIGHT:

SCALE:1:2

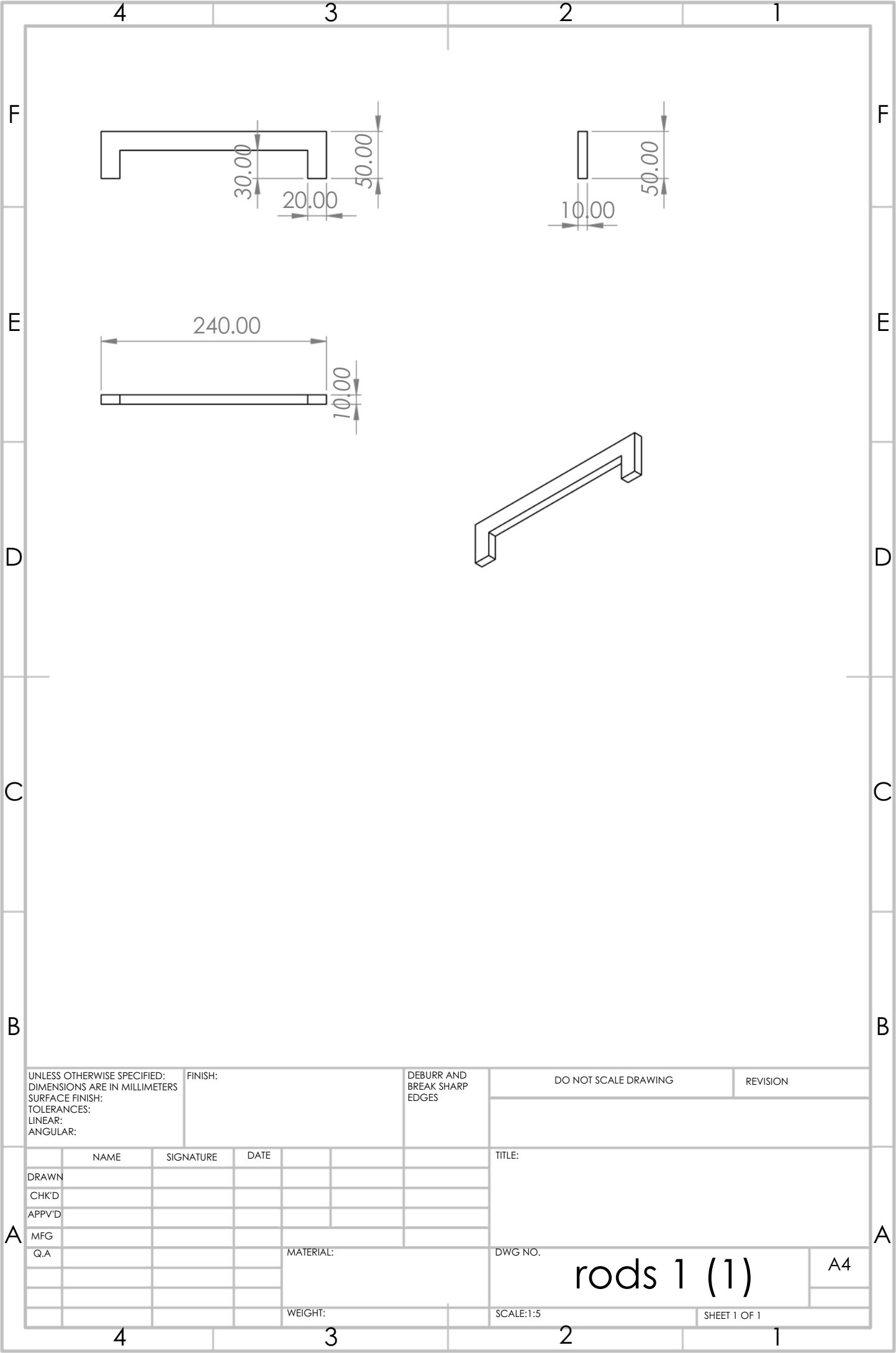
SHEET 1 OF 1

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BREAK SHARP
EDGES

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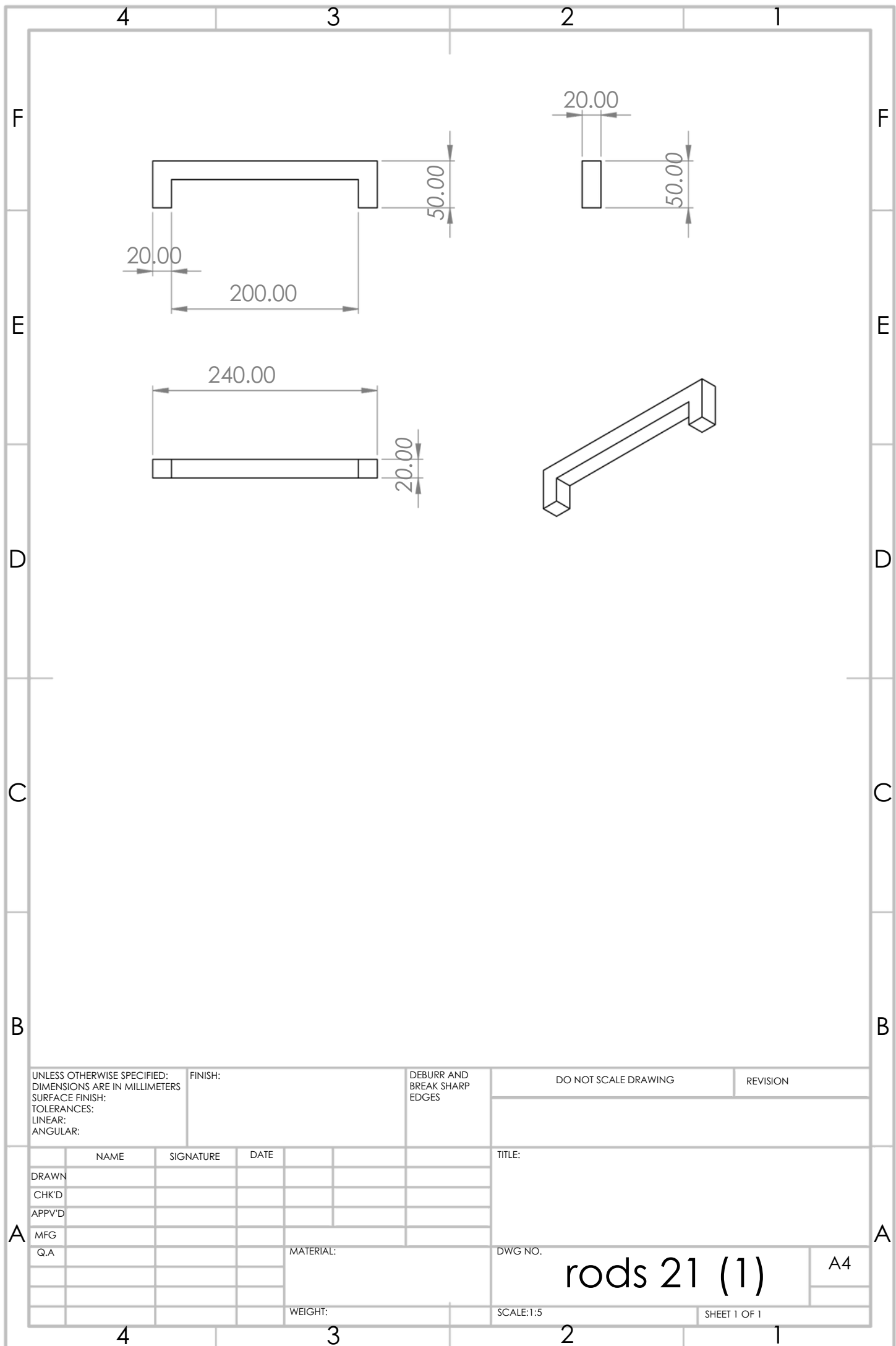
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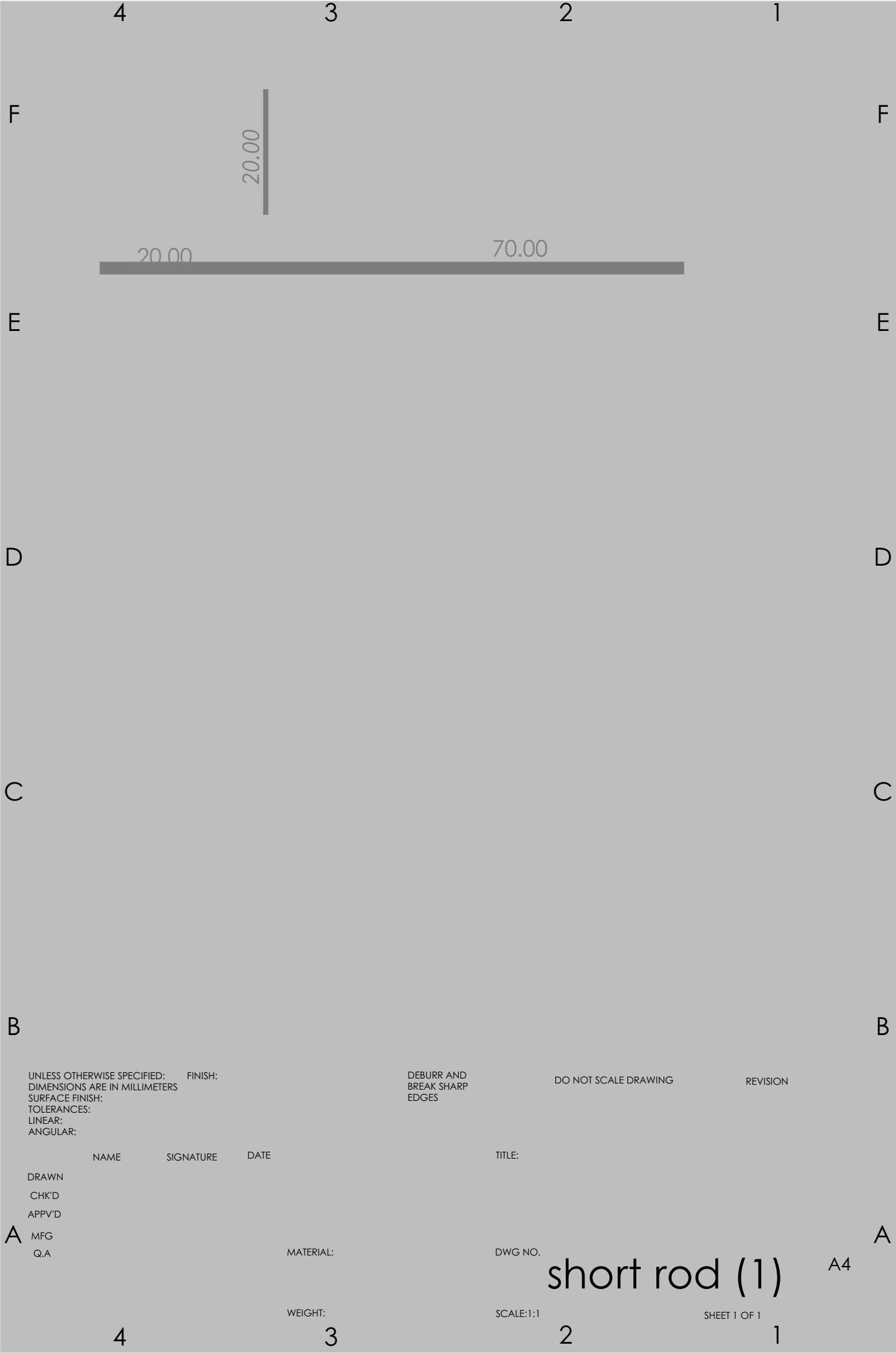
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SHEET 1 OF 1

rods 1 (1)

A4





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ANGULAR:

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BREAK SHARP
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REVISION

NAME SIGNATURE DATE

TITLE:

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APPV'D

A MFG
Q.A

MATERIAL:

DWG NO.

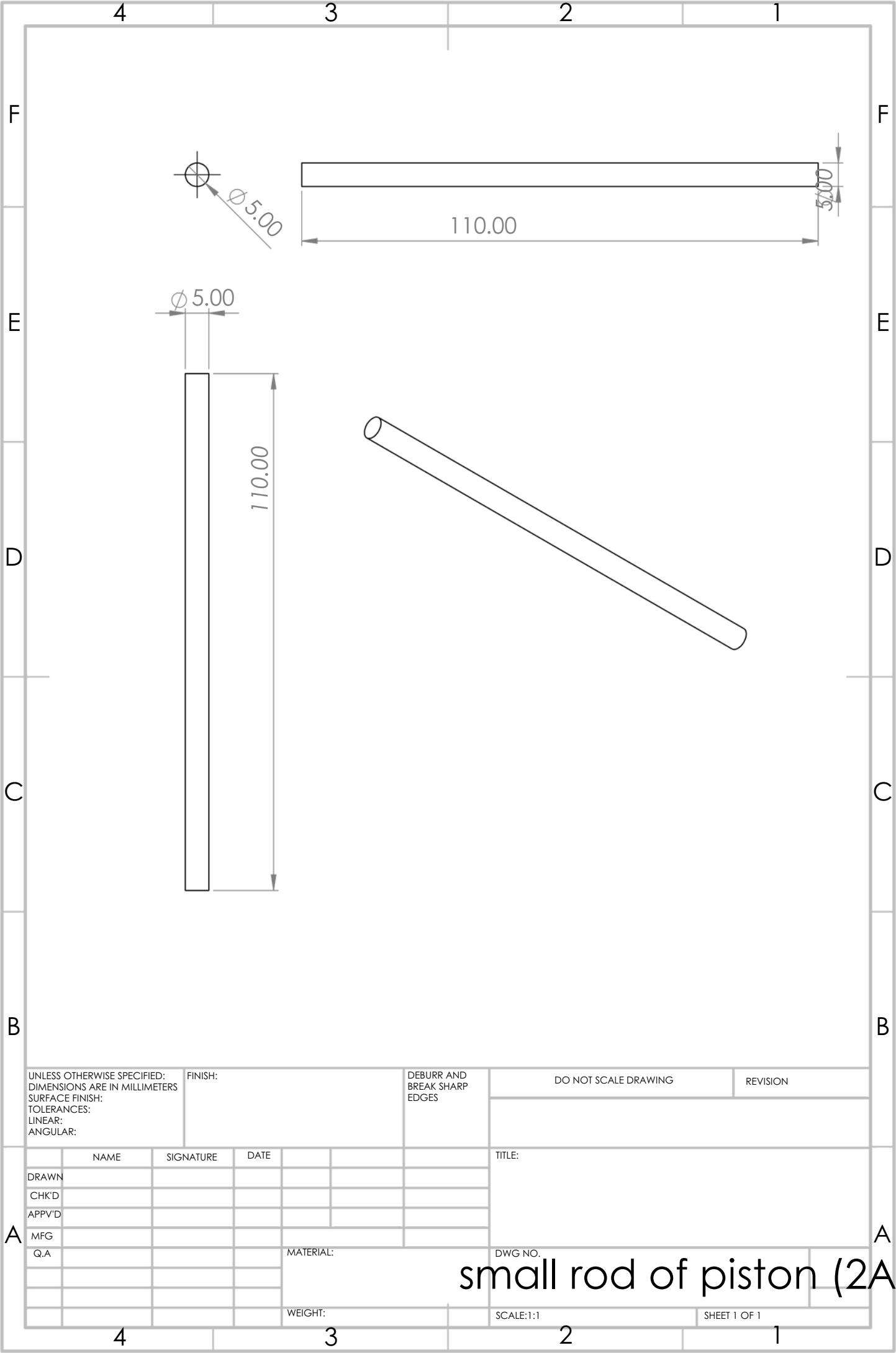
short rod (1)

A4

WEIGHT:

SCALE:1:1

SHEET 1 OF 1



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TOLERANCES:
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BREAK SHARP
EDGES

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MATERIAL:

WEIGHT:

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DWG NO.

SCALE:1:1

SHEET 1 OF 1

small rod of piston (2A4)